Practice Set 6 Paper 1 – Basic Sciences in Urology

1. Q1. Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma
2. Q2. Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities.
3. Q3a. Premalignant lesion of penis
4. Q3b. EPN. Prognostic factors & conserv. mgt.
5. Q3c. VCUG
6. Q3d. Criteria for evaluation of quaLliy of medical research papers
7. Q3e. Mgt. of acute hyperkalemia.
8. Q3f. Oncogene

Practice paper compiled from the question bank.

Q1. Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma

Type: LAQ (25 marks) (pre-operative management also answered in 2020-Summer Paper 2, Q3c)

Assumed meaning: "arenal" = adrenal glands.

A. Anatomy

Gross

- Paired retroperitoneal glands on the upper renal poles (11th–12th ribs), inside Gerota fascia, separated from the kidney by thin connective tissue. Yellow-orange, 4–6 × 2–3 cm, about 5 g each.

Feature	Right	Left
Shape, level	Triangular; higher	Crescentic; lower (to L1)
Relations	Liver anteriorly, duodenum anteromedially, IVC (inferior vena cava) medially, diaphragmatic crus behind	Pancreas body and splenic vessels anteriorly, aorta medially, crus behind
Vein	**Short (<1 cm), into posterior IVC**: tears easily	**Longer, into left renal vein**, often joined by inferior phrenic vein

Blood supply

- **"Many arteries in, one vein out."** Superior adrenal (from inferior phrenic), middle (aorta), inferior (renal artery) arteries break into 10–50 small branches.
- Cortical sinusoids drain into the medulla, so cortisol-rich blood reaches the chromaffin cells. **Cortisol induces PNMT (phenylethanolamine-N-methyltransferase)**, which is why the medulla makes adrenaline.

Nerves: preganglionic sympathetic fibres (T11–L2, via splanchnic nerves, no synapse in coeliac ganglion) end directly on chromaffin cells. The medulla is a modified sympathetic ganglion.

Embryology: cortex from coelomic mesoderm (adrenogonadal ridge, weeks 3–4); medulla from neural crest. Hence adrenals stay orthotopic in renal agenesis, and adrenal rests occur along the path of gonadal descent.

Histology (zones from outside in: glomerulosa, fasciculata, reticularis = "salt, sugar, sex")

Zone	Share	Product	Main control
Glomerulosa	~15% of cortex	Aldosterone	Angiotensin II, potassium
Fasciculata	~80%	Cortisol (up to 20 mg/day)	ACTH (adrenocorticotropic hormone)
Reticularis	5–7%	DHEA (dehydroepiandrosterone), DHEA-S, androstenedione	ACTH
Medulla	~10% of gland	Adrenaline 80%, noradrenaline 19%, dopamine 1%	Sympathetic nerves

B. Steps in hormone production

1. Steroids (cortex)

- Cholesterol (mainly from LDL [low-density lipoprotein]) is carried into mitochondria by **StAR (steroidogenic acute regulatory protein)**: the rate-limiting step, driven by ACTH.
- **Side-chain cleavage enzyme (CYP11A1)** converts it to **pregnenolone**. The enzymes present in each zone then decide the product:

```
Pregnenolone -3βHSD-> Progesterone -21OH-> 11-deoxycorticosterone -11βOH-> Corticosterone -
aldosterone synthase (CYP11B2)-> ALDOSTERONE (glomerulosa only)
Pregnenolone -17αOH-> 17-OH pregnenolone/progesterone -21OH-> 11-deoxycortisol -11βOH->
CORTISOL (fasciculata)
17-OH pregnenolone -17,20-lyase-> DHEA -> androstenedione (reticularis)
```

- 3βHSD (3β-hydroxysteroid dehydrogenase). Steroids are not stored; they diffuse out and act on intracellular receptors.
- **Regulation:** CRH (corticotropin-releasing hormone) → ACTH → cortisol, with negative feedback and morning peak; RAAS (renin–angiotensin–aldosterone system) and potassium for aldosterone.
- **Clinical link:** 21-hydroxylase deficiency is the commonest cause of CAH (congenital adrenal hyperplasia).

2. Catecholamines (medulla)

```
Tyrosine -tyrosine hydroxylase (rate-limiting)-> L-DOPA -decarboxylase-> Dopamine
-dopamine β-hydroxylase-> Noradrenaline -PNMT (cortisol-induced)-> Adrenaline
```

- Stored in vesicles, released by exocytosis. **COMT (catechol-O-methyltransferase)** inside chromaffin cells forms **metanephrine and normetanephrine** continuously, which is why metanephrines are the best test. MAO (monoamine oxidase) and liver form VMA (vanillylmandelic acid).

C. Perioperative management of phaeochromocytoma

Why: handling the tumour causes catecholamine surges (crisis, arrhythmia); vein ligation causes sudden withdrawal (hypotension, hypoglycaemia). Mortality fell from up to 50% to <3% with blockade.

Pre-operative (7–14 days)

- Confirm with plasma free or urinary fractionated metanephrines; CT (computed tomography) or MRI; functional imaging (^{68}Ga-DOTATATE PET [positron emission tomography], MIBG [metaiodobenzylguanidine]) if extra-adrenal or hereditary; **never biopsy**; genetic testing (about one-third hereditary).
- ECG (electrocardiogram) and echocardiography for catecholamine cardiomyopathy.

Drug	Use
Phenoxybenzamine (irreversible non-selective α)	First line: 10 mg twice daily, increased every 2–3 days; about 1 mg/kg total
Doxazosin, prazosin (selective α1)	Fewer side effects
Calcium channel blockers	Alternative strategy or add-on
β1-blocker (atenolol, metoprolol)	**Only after α-blockade**, for tachycardia
Metyrosine	Blocks tyrosine hydroxylase; add-on; sedation, extrapyramidal effects

- **Never β before α:** losing β2 vasodilatation leaves α-vasoconstriction unopposed: crisis.
- **Target:** seated BP 120–130/80 mmHg with mild postural drop.

- **Volume expansion:** high-salt diet and fluids; 1–2 L IV (intravenous) fluid the night before. Last phenoxybenzamine dose the night before surgery.

Intra-operative

- Arterial and central lines; laparoscopic adrenalectomy (open for large or invasive tumours); minimal handling; **early adrenal vein ligation** after warning the anaesthetist.
- Surges: nitroprusside, nicardipine, phentolamine. Tachyarrhythmia: esmolol, lidocaine. Post-ligation hypotension: fluids ± vasopressors.
- Cortical-sparing adrenalectomy in MEN2 (multiple endocrine neoplasia type 2) and VHL (von Hippel–Lindau).

Post-operative

- HDU (high dependency unit): hypotension, **rebound hypoglycaemia** (insulin no longer suppressed), steroid cover if bilateral.
- Metanephrines after surgery, then **yearly for life** (recurrence up to 16%).

Diagram to draw: posterior view of both adrenals with three arteries and the two veins.

Key points to remember:

- Right vein short into IVC; left vein into left renal vein.
- StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
- Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
- α-blockade 7–14 days, β only after α, volume expansion; watch glucose.

Q2. Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities.

Type: LAQ (25 marks) (bladder embryology also answered in 2021-Summer Paper 3, Q1)

A. Embryology of the bladder

1. Division of the cloaca (weeks 5–6)

- The **urorectal septum** (Tourneux and Rathke folds) grows caudally and divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. It meets the cloacal membrane, which perforates to form the urogenital and anal openings.
- The **entry of the Wolffian (mesonephric) ducts** divides the UGS:
  - **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
  - **Caudal (definitive) UGS** → phallic urethra in males; vestibule and distal vagina in females.

2. Wall layers

- Urothelium from UGS endoderm. Lamina propria, detrusor and adventitia from surrounding splanchnic mesenchyme, by epithelial–mesenchymal signalling.

3. Trigone (weeks 5–6)

- The common excretory ducts dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the Wolffian ducts.
- Ureteric orifices migrate **cranially and laterally**; Wolffian ducts end as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric. Current view (mouse lineage tracing): **mostly bladder smooth muscle**.
- **Weigert–Meyer:** bud too low → lateral orifice → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Urachus: the bladder descends into the pelvis (months 4–5); the allantois obliterates into the **urachus** (median umbilical ligament).

5. Function: urine from about week 10; after week 16 amniotic fluid is mainly fetal urine.

B. Exstrophy–epispadias complex (EEC): theories

- **Spectrum:** glanular epispadias → classic bladder exstrophy (CBE, >50%; about 1 in 46,000; male:female 2.3:1) → cloacal exstrophy.
- **Core defect:** the **cloacal membrane is not reinforced by mesodermal ingrowth**. This mesoderm normally forms the lower abdominal muscles and pelvic bones, so the thin membrane ruptures. **Timing and extent of rupture** decide the variant.

Theory	Proposed defect	Result
Cloacal membrane overdevelopment (Marshall–Muecke)	Abnormally large membrane in week 4 blocks medial mesenchymal migration	Rupture before septum descent → cloacal exstrophy; after → bladder exstrophy; limited → epispadias
Lateral body-wall fold failure	Folds fail to meet in midline	Abdomen and pelvis → cloacal; pelvis alone → classic
Caudal genital hillocks	Genital tubercles fuse **below** rather than above the membrane	Dorsal urethral defect (epispadias)

Theory	Proposed defect	Result
Abnormal body-stalk insertion	Caudal insertion prevents mesenchyme interposition	Superficial unstable membrane disintegrates
Bony pelvis first	Pelvic ring primordium fails to rotate	Pelvic structures cannot meet → bladder herniates

- All theories converge on **absent infraumbilical midline mesoderm**.
- **Also know (aetiology):** multifactorial; recurrence in offspring of an affected parent 1 in 70; 22q11.2 duplication (commonest variant), ISL1 susceptibility gene, p63 under-expression; IVF (in vitro fertilisation) and high-dose early progesterone raise risk; maternal smoking linked to cloacal exstrophy.

C. Skeletal abnormalities

1. Classic bladder exstrophy (3D [three-dimensional] CT [computed tomography] studies)

- **Rotational:** posterior pelvis externally rotated (mean 12° each side); anterior segment externally rotated (mean 18° each side); sacroiliac joint rotated; acetabular and femoral retroversion.
- **Dimensional: pubic diastasis** (mean 4.8 cm); **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%; posterior segment length normal; larger sacrum.
- Bone histology is normal, so growth potential is normal.
- **Clinical effects:** short pendular penis (anterior segment short and splayed); wide hips, waddling gait and out-toeing in early childhood. These cause little disability and improve with growth.
- **Pelvic floor:** levator ani displaced posteriorly, flattened; anterior anus; later pelvic organ prolapse.
- **Spine:** minor variants (spina bifida occulta, lumbarisation, sacralisation) fairly common; scoliosis and dysraphism uncommon.

2. Cloacal exstrophy

- Widest, **asymmetric** diastasis (about twice CBE), sacroiliac malformation, hip dislocation.
- **Spinal anomalies in 85–100%** (lumbar myelodysplasia commonest; lipomeningocele); MRI (magnetic resonance imaging) for all.
- Limb defects: **clubfoot** (commonest), limb shortening, tibial and fibular deformity.

3. Isolated epispadias: male symphysis widened (less than CBE); female symphysis usually closed or narrow.

Surgical significance: pelvic osteotomy

- Closing the pelvic ring places the bladder neck deep in the pelvis, reduces tension and helps continence.
- **Indications:** diastasis >4 cm or poorly malleable pelvis; closure after 72 hours; failed closure; all cloacal exstrophy. Under 72 hours with malleable pubes: closure without osteotomy and traction.
- **Combined bilateral anterior innominate + vertical iliac osteotomy** (most used) with external fixator.
- Epispadias repair and continence surgery: see Practice Set 6 Paper 3, Q2.

Diagram to draw: cloacal membrane with urorectal septum and failed mesodermal ingrowth; exstrophy pelvis with diastasis.

Key points to remember:

- Vesicourethral canal → bladder; urachus from allantois.
- Marshall–Muecke: overdeveloped cloacal membrane; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis ~4.8 cm.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.

Q3a. Premalignant lesion of penis

Type: SAQ (10 marks) (premalignant conditions also answered in 2021-Summer Paper 1, Q3a)

Definition

- A lesion with raised risk of invasive SCC (squamous cell carcinoma). The key entity is **PeIN (penile intraepithelial neoplasia)**: dysplastic squamous epithelium with an **intact basement membrane**.

Classification

1. PeIN (HPV-associated or HPV-independent)

- **HPV (human papillomavirus)-associated** (commonest; basaloid and warty): viral E6 inactivates p53, E7 inactivates Rb (retinoblastoma protein); **p16** is the marker.
- **HPV-independent (differentiated):** arises in lichen sclerosus.
- **WHO (World Health Organization) 2022:** all PeIN is high grade; grades I–III abandoned.

Clinical entity	Site, appearance	Risk of invasion
Erythroplasia of Queyrat	Glans or prepuce; red velvety plaque	**10–33%**
Bowen disease	Shaft; scaly red plaque	About 5%
Bowenoid papulosis	Shaft papules, young men; HPV 16	Does not progress; may regress

2. Other lesions

- **Lichen sclerosus (BXO [balanitis xerotica obliterans]):** SCC in **2.3–9%** (HPV-independent type).
- **Buschke–Löwenstein tumour** (giant condyloma; verrucous carcinoma): HPV 6/11; locally destructive, rarely metastasises; may harbour SCC.
- **Also know:** PEKMB (pseudoepitheliomatous keratotic and micaceous balanitis, controversial); condyloma acuminatum; chronic phimosis; cutaneous horn; Kaposi sarcoma (HHV-8 [human herpesvirus 8], check HIV [human immunodeficiency virus]).

Evaluation

- **Multiple biopsies of adequate depth** to exclude invasion; p16 staining. Examine inguinal nodes; partner screening for cervical neoplasia.

Treatment (organ-preserving)

Situation	Treatment
Prepuce	Circumcision or excision with clear margin
Glans	**Topical 5-FU (5-fluorouracil) or 5% imiquimod**; **laser** (Nd:YAG [neodymium:yttrium-aluminium-garnet] preferred; CO_2 too superficial); **glans resurfacing** with skin graft
Shaft	Excision; Mohs surgery
Lichen sclerosus	Clobetasol 0.05% for 3 months then maintenance; circumcision
Buschke–Löwenstein	Wide excision; avoid radiotherapy (anaplastic change)

- Long-term follow-up (recurrence common after organ-preserving treatment). **Prevention:** neonatal circumcision, HPV vaccination, avoiding smoking.

Diagram to draw: epithelium with full-thickness atypia above an intact basement membrane.

Key points to remember:

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33%, Bowen about 5%, bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.

Q3b. EPN. Prognostic factors & conserv. mgt.

Type: SAQ (10 marks) (prognostic indicators also answered in 2024-Winter Paper 3, Q3c)

Assumed meaning: EPN = emphysematous pyelonephritis.

Definition

- Acute **necrotising** infection of renal parenchyma and perirenal tissue by **gas-forming organisms** (*E. coli*, *Klebsiella*, *Proteus*). Mostly **diabetics**; obstruction in about a quarter. Mortality 19–43%.
- **Mechanism:** high tissue glucose fermented to CO_2 in poorly perfused tissue.
- **CT (computed tomography)** is the imaging of choice.

Prognostic factors

1. Huang–Tseng CT classification (2000)

Class	Gas location
1	Collecting system only
2	Parenchyma only
3A	Perinephric space
3B	Pararenal space
4	Bilateral or solitary kidney

2. Wan CT pattern

- **Type I:** streaky gas, **no fluid**: mortality 50–60%.
- **Type II:** fluid with bubbly or loculated gas: <20%. Fluid means a contained collection that can be drained.

3. Clinical risk factors (Huang–Tseng): thrombocytopenia, AKI (acute kidney injury), altered sensorium, shock.

- **Also adverse:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

Conservative management (now standard)

1. Resuscitation: fluids, inotropes, ICU (intensive care unit) if shocked; **tight glucose control**; correct electrolytes.

2. Antibiotics: broad-spectrum IV (intravenous), adjusted to culture; usually several weeks in total.

3. Drainage

- **Relieve obstruction:** DJ (double-J) stent or PCN (percutaneous nephrostomy).
- **PCD (percutaneous catheter drainage)** of gas and fluid collections under CT or ultrasound; multiple drains if needed; repeat CT to check.

Algorithm

```
All: resuscitation + glucose control + IV antibiotics + relieve obstruction
Class 1–2                      -> antibiotics ± PCD
Class 3–4, <2 risk factors     -> antibiotics + PCD
Class 3–4, >=2 risk factors,
or PCD failure                 -> nephrectomy (emergency or delayed)
```

- Emergency nephrectomy carried 40–50% mortality, so drainage-first care replaced it. Delayed nephrectomy for a non-functioning kidney on renal scan.

- **Emphysematous pyelitis** (class 1) often responds to antibiotics alone.

Diagram to draw: CT sketches of Huang–Tseng classes.

> **Key points to remember:**
> - Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral/solitary.
> - Risk factors: thrombocytopenia, AKI, altered sensorium, shock.
> - Wan type I (no fluid) 50–60% mortality; drainage first, nephrectomy for failure.

Q3c. VCUG

Type: SAQ (10 marks)

Definition

- **VCUG (voiding cystourethrogram; MCU, micturating cystourethrogram):** fluoroscopic study of bladder filling and voiding after catheter instillation of contrast. It shows the bladder, **posterior urethra** and UVJ (ureterovesical junction) competence.
- **Gold standard for VUR (vesicoureteral reflux)** and PUV (posterior urethral valves).

Indications

- **Children:** febrile UTI (urinary tract infection) with abnormal ultrasound or recurrent infection (EAU [European Association of Urology]: also age <12 months); bilateral hydroureteronephrosis (PUV); dilated ureter; ureterocele; neurogenic bladder.
- **Adults:** stricture (with RGU [retrograde urethrogram]), diverticula, fistula, post-operative leak, voiding dysfunction.

Technique

- **Sterile urine** first. After UTI, wait until urine is sterile and the child is improving (about a week); the old 4–6-week delay is unnecessary.
- Scout film → catheter: **5 Fr feeding tube (infants), 6–8 Fr (children), 14–16 Fr (adults)** → confirm tip position.
- **Gravity fill** (≤70 cm H_2O) to expected capacity: <2 years weight (kg) × 7; 2–14 years (age × 30) + 30 mL; >14 years 300–500 mL. (Koff formula: (age + 2) × 30 mL.)
- **Oblique views** (UVJ, prostatic urethra) → remove catheter → **void under fluoroscopy** → **post-void film**.
- **Cyclic VCUG** (2–3 fill–void cycles) detects 12–20% more VUR.

Findings

- **VUR grade (International Reflux Study, 1981):**

Grade	Appearance
I	Ureter only, not dilated
II	Pelvis and calyces, no dilatation
III	Mild–moderate dilatation, minimal blunting
IV	Moderate dilatation and tortuosity
V	Gross dilatation; papillary impressions lost

- Filling-phase reflux has a poorer resolution prognosis than voiding-only reflux.
- **PUV:** dilated, elongated posterior urethra, prominent bladder neck, trabeculation.
- Ureterocele, diverticula, residual urine; post-void pelvic retention suggests PUJ (pelviureteric junction) obstruction.

Limitations and complications

- Invasive, radiation, UTI, catheter trauma; snapshot (occult VUR); interobserver variability; autonomic dysreflexia (injury above T6).

Alternatives

- **Direct radionuclide cystography:** more sensitive, lower dose, poor anatomy.

- **CeVUS (contrast-enhanced voiding urosonography):** no radiation, accuracy close to VCUG.
- **PIC (positional instillation of contrast) cystography:** occult VUR with negative VCUG.

Diagram to draw: VUR grades I–V.

> **Key points to remember:**
> - Gold standard for VUR and PUV; sterile urine first.
> - Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
> - Grades I–II non-dilating, III–V dilating.

Q3d. Criteria for evaluation of quaLliy of medical research papers

Type: SAQ (10 marks) (evidence-based medicine also answered in 2020-Summer Paper 4, Q3f)

Assumed meaning: "quaLliy" = quality, i.e. critical appraisal.

Definition

- **Critical appraisal:** systematic assessment of a paper's **validity, results and applicability**; the "Appraise" step of EBM (evidence-based medicine).

Three core questions

1. Is it valid? (internal validity)

- **PICO question** (Population, Intervention, Comparison, Outcome); suitable design.
- **RCTs (randomised controlled trials):** true randomisation, **allocation concealment**, comparable groups at baseline, **blinding**, complete follow-up (>80%), **ITT (intention-to-treat) analysis**.
- **Bias** checked:

Bias	Meaning
Selection	Groups differ at entry
Confounding	Third factor (comorbidity) linked to exposure and outcome
Performance, detection, attrition	Unequal care, outcome assessment or dropouts
Lead-time, length-time; stage migration (Will Rogers)	Screening or better tests make survival look longer
Publication	Negative studies unpublished

2. What are the results?

- Effect: RR (relative risk), **ARR (absolute risk reduction), NNT (number needed to treat) = 1/ARR**.
- Precision: **95% CI (confidence interval)**; p value; **sample size and power** (usually 80%; α 0.05).
- Surrogate end points must meet the **Prentice criteria**; validated outcome tools; Clavien–Dindo grading.

3. Does it apply? (external validity)

- Similar patients and setting; benefit outweighs harm.

Levels of evidence (Oxford CEBM [Centre for Evidence-Based Medicine], as used by EAU [European Association of Urology])

Level	Design
1a / 1b	Systematic review of RCTs / single RCT
2a / 2b	Controlled non-randomised / cohort study
3	Case-control, case series
4	Expert opinion

- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; strong or weak.

Reporting checklists

Design	Checklist
RCT	**CONSORT** (Consolidated Standards of Reporting Trials)
Systematic review / meta-analysis	**PRISMA** (Preferred Reporting Items for Systematic reviews and Meta-Analyses)
Observational	**STROBE** (Strengthening the Reporting of Observational studies in Epidemiology)
Diagnostic accuracy	**STARD** (Standards for Reporting Diagnostic accuracy studies)
Surgical innovation	**IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study)

- **Also know:** Cochrane RoB 2 (risk-of-bias) tool; trial registration; conflicts of interest.

Diagram to draw: evidence pyramid.

> **Key points to remember:**
> - Three questions: valid, results, applicable.
> - Randomisation, concealment, blinding, ITT, follow-up >80%.
> - CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Q3e. Mgt. of acute hyperkalemia.

Type: SAQ (10 marks) (also answered in 2025-Winter Paper 1, Q3e)

Definition and physiology

- Serum potassium **>5.5 mmol/L**; mild 5.5–5.9, moderate 6.0–6.4, **severe ≥6.5** (UK Kidney Association).
- 98% of potassium is intracellular; the ratio sets the resting membrane potential, so high levels cause **arrhythmia and cardiac arrest**.
- Defence: shift into cells (insulin, β_2-agonists via Na^+/K^+-ATPase), then renal excretion (distal sodium delivery, urine flow, aldosterone).

Causes

Mechanism	Examples
Pseudohyperkalaemia	Haemolysis, tourniquet, fist clenching, thrombocytosis
Reduced excretion	AKI (acute kidney injury), CKD (chronic kidney disease), **obstructive uropathy** (type 4 RTA [renal tubular acidosis]), adrenal insufficiency; drugs: ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers), spironolactone, NSAIDs (non-steroidal anti-inflammatory drugs), tacrolimus
Shift out of cells	Acidosis, insulin deficiency, β-blockers
Cell release	**Rhabdomyolysis** (prolonged lithotomy or flank position), tumour lysis, crush
Urological	**Jejunal conduit** (reabsorbs urinary potassium)

Assessment

- Repeat the sample to **exclude pseudohyperkalaemia** if the ECG (electrocardiogram) is normal.
- **ECG:** peaked T (earliest) → prolonged PR, wide QRS → loss of P → sine wave → VF (ventricular fibrillation).
- Creatinine, blood gas, CK (creatine kinase); stop potassium sources.

Emergency treatment (in order)

Step	Drug and dose	Onset and effect
1. Protect the heart	**Calcium gluconate 10%, 10 mL IV (intravenous)** over 5–10 min; repeat if ECG unchanged	1–3 min; does not lower potassium
2. Shift into cells	**Insulin 10 units regular IV with 50 mL of 50% dextrose**	15–30 min; lowers about 0.6–1 mmol/L; monitor glucose
	Salbutamol 20 mg nebulised	Additive
	Sodium bicarbonate	Only with metabolic acidosis
3. Remove	Loop diuretic if passing urine; **binders**: sodium zirconium cyclosilicate (about 1 h), patiromer (about 7 h), sodium polystyrene sulfonate (slow; colonic necrosis)	Hours
	Haemodialysis	Refractory or oliguric

- **Relieve obstruction** (catheter, nephrostomy); watch for post-obstructive diuresis.
- Monitor ECG, potassium and glucose; treat the cause.

Diagram to draw: flowchart: ECG change or K^+ ≥6.5 → calcium → shift → remove.

Key points to remember:

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

Q3f. Oncogene

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 1, Q3f)

Definition

- **Proto-oncogene:** normal gene that promotes cell growth and division (an "accelerator").
- **Oncogene:** its mutated or overactive form, driving uncontrolled proliferation. The change is **gain of function** and **dominant**: one altered allele is enough.
- The first human oncogene, **HRAS**, was found in the **T24 bladder cancer** cell line.

Mechanisms of activation

Mechanism	Effect	Example
Activating point mutation	Protein permanently "on"	RAS; FGFR3 (fibroblast growth factor receptor 3) in low-grade papillary bladder cancer; germline MET in hereditary papillary RCC (renal cell carcinoma)
Gene amplification	More copies → overexpression	MYC (prostate, bladder); AR (androgen receptor) in castration-resistant prostate cancer; HER2 (human epidermal growth factor receptor 2)
Translocation / fusion	Gene under a strong promoter, or fusion protein	**TMPRSS2–ERG** (about half of prostate cancers); TFE3 fusions (translocation RCC)

- **Also know:** promoter hypomethylation can switch oncogenes on.

Functional classes

Class	Examples
Growth factors	PDGF (platelet-derived growth factor)
Receptor tyrosine kinases	EGFR (epidermal growth factor receptor), HER2, FGFR3, MET, KIT
Signal transducers	**RAS** (GTP [guanosine triphosphate]-binding; mutation locks it on), BRAF, PI3K
Transcription factors	**MYC**, ERG
Cell cycle / apoptosis	Cyclin D1, BCL-2

Oncogene versus tumour suppressor gene

	Oncogene	Tumour suppressor
Role	Accelerator	Brake, DNA repair
Change	Gain of function	Loss of function
Alleles	One	Usually both (**Knudson two-hit**)
GU (genitourinary) examples	MYC, MET, FGFR3, ERG	RB1, TP53, VHL, PTEN

Urological significance

- **Bladder:** FGFR3 and TERT promoter mutations in >60% of low-grade papillary tumours; HRAS mutually exclusive with FGFR3; MYC overexpressed in over half.
- **Kidney:** MET activation in papillary RCC.
- **Prostate:** TMPRSS2–ERG, MYC (8q gain), AR amplification under hormone therapy.

Targeted therapy ("oncogene addiction")

- **Erdafitinib** (FGFR inhibitor) for FGFR3/2-altered metastatic urothelial cancer; MET-active TKIs (tyrosine kinase inhibitors) in papillary RCC; AR-targeted drugs in prostate cancer.

Diagram to draw: growth factor → receptor → RAS → RAF → MEK → ERK → MYC.

Key points to remember:

- Gain of function, one allele; mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

Rapid Revision Sheet – Practice Set 6 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma (LAQ)

- Right vein short into IVC; left vein into left renal vein.
- StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
- Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
- α-blockade 7–14 days, β only after α, volume expansion; watch glucose.

Q2. Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities. (LAQ)

- Vesicourethral canal → bladder; urachus from allantois.
- Marshall–Muecke: overdeveloped cloacal membrane; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis ~4.8 cm.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.

Q3a. Premalignant lesion of penis (SAQ)

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33%, Bowen about 5%, bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.

Q3b. EPN. Prognostic factors & conserv. mgt. (SAQ)

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral/solitary.
- Risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; drainage first, nephrectomy for failure.

Q3c. VCUG (SAQ)

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

Q3d. Criteria for evaluation of quaLliy of medical research papers (SAQ)

- Three questions: valid, results, applicable.
- Randomisation, concealment, blinding, ITT, follow-up >80%.
- CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

Q3e. Mgt. of acute hyperkalemia. (SAQ)

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

Q3f. Oncogene (SAQ)

- Gain of function, one allele; mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.